May 30, 2025

Dear Dr. Ferguson,

We are resubmitting our manuscript (XLM-2025-0564), entitled "Compensation in audiovisual speech perception: discounting the pen in the mouth." We are grateful to you and the reviewers for your highly constructive feedback on our initial submission. We were heartened to see that the reviewers "found the study to be interesting, well-executed and potentially impactful for understanding audiovisual speech perception" and, in particular, that reviewers considered the focus on non-linguistic visual information influencing speech perception clearly motivated and valuable.

We very much appreciate the feedback from reviewers, which helped us address the presentational points they raised. Specifically:

1. <u>The revised introduction</u> now clarifies our contributions to previous work that has used similar audiovisual manipulations as part of experiments on lexically-guided perceptual recalibration (especially, Kraljic et al. 2008, as well as Kraljic & Samuel, 2011; Liu & Jaeger, 2018).

   In short, these studies investigated how a visually-presented pen in the mouth (vs. hand) of a speaker during the lexically-guided exposure phase affected listeners perception during a subsequent *audio-only* test phase. The thought-provoking finding of these studies was that perceptual recalibration can be blocked or reduced if the audio-visual exposure items with shifted pronunciations occur only while the speaker has a pen in the mouth. Competing explanations have been offered for this reduction of perceptual recalibration—from causal inferences (e.g., Kraljic et al., 2008; Liu & Jaeger, 2018) to separate exemplar storage of audio-only vs. audio-visual inputs (Kraljic & Samuel, 2011).

   Critically, the design of these studies did not allow researchers to estimate how the placement of the pen affects perception *in that moment*: the pen was only ever displayed during the *exposure* phase, which used a lexical decision—rather than categorization—task. This makes it difficult to adjudicate between the competing hypotheses about the pen's downstream effects during the test phase. The revised introduction now clarified that this is the knowledge gap addressed by our study, which is the first study to use audiovisual test continua that manipulate the location of the pen (mouth vs. hand). We have also revised the discussion to spell out in more detail what our results mean for this earlier line of work.

2. <u>The revised methods</u> now contain all the information requested by reviewers (moved there from the supplementary information). We agree that this makes the manuscript more accessible. The reviewers' comments also made us realize that we had made a small mistake in reporting the exclusion criteria. This is now fixed, and it didn't change any of the results.

3. <u>The revised discussion of Study 2</u> now transparently acknowledges the concern raised by Reviewer 1. In that context, we now also elaborate on participant responses to the post-experiment survey, and on Study 2b (described in detail in the SI). These both ameliorate—but do not fully remove—that concern. This revision, we hope, helps readers to maintain an adequate level of uncertainty about our interpretation of our results.

Below we provide a point-by-point response to the reviewers' comments; original comments are bounded by boxes with our response immediately below each box. We have also fixed the typos pointed out by all reviewers; we thank them for their attention and have given the manuscript another careful read ourselves to eliminate further mistakes.

Sincerely,

Shawn Cummings
Menghan Yang
Gevher Karboga
T. Florian Jaeger

Reviewer 1

[summary omitted] Overall, this is a very nice study that addresses an interesting question about speech perception. The literature review is brief, but sufficient to give an quick overview of the theoretical background. The research questions are clearly spelled-out, the writing is quite clear, and the narrative is easy to follow.

This work has many positive features, however there a couple of important issues that prevent me from recommending it for publication in its current form.

Thank you for this careful summary and contextualization of our manuscript, as well as for your positive assessment.

It is not entirely clear to me what exactly is the novelty of this work. As mentioned in the Discussion, the question of how listeners deal with shifted speech input as a function of whether the shift is attributed to idiosyncratic characteristics of the speaker (e.g., accent) versus incidental factors (e.g., pen in mouth) has already been addressed in the past. Kraljic et al (2008) asked this question and found that the effect of interest (perceptual recalibration in that case) goes away when the shift is attributed to incidental factors. Thus, the main conclusion of that study was the same as here: listeners 'explain away' the shift when it can be attributed to an incidental factor. I understand that there are some novel aspects introduced in this work, but they are not sufficiently highlighted/clarified right now. One suggestion would be to first present this work in the Introduction (rather than the Discussion) and explain what the present study adds that is new / different from that work.

We thank the reviewer for highlighting this point. Indeed, our experiments were in part motivated by an effort to better understand the results of Kraljic and colleagues' seminal work, and to take some steps towards disentangling competing explanations for the result obtained by Kraljic et al. (2008).

For instance, Kraljic & Samuel (2011) proposed an alternative explanation that does not appeal to 'explaining away', but rather appeals to specifics of exemplar storage. Liu & Jaeger (2018) argued that this alternative account faces both theoretical and empirical issues. In a series of experiments, Liu & Jaeger (2018) argued for an elaborated version of the original proposal in Kraljic et al. (2008): that the effects of pen placement are rooted in causal inferences under uncertainty—uncertainty about whether the pen or the talker is the cause for the observed unexpected pronunciation. In short, competing explanations have been offered for the findings of Kraljic et al. (2008) and subsequent related works.

Critically, none of these earlier experiments addressed how the pen actually affects perception *in the moment*. Instead, all previous experiments assessed effects during audio-only test phases that followed the audio-visual exposure phase in which pen placement was manipulated. In those original experiments, it is thus unclear (1) how the pen affects perception (as opposed to subsequent perceptual recalibration) and (2) what assumptions listeners held about the pen placement during test (where no audiovisual materials were used).

These knowledge gaps—which we address in the present work—have limited researchers' ability to distinguish between the competing hypotheses described above. Additionally, our experiments have the potential to shed light on some more nuanced complexities in the line of work inspired by Kraljic and colleagues. For example, Liu & Jaeger (2018) found that the blocking/reduction of perceptual recalibration in the pen-in-the-mouth condition is asymmetric between "s" and "sh"-biased exposure. In the revised general discussion, we clarify how our findings offer the beginning of a new explanation for this asymmetry.

I really liked the idea of testing whether the visibility of the pen-in-the-mouth manipulation modulates the effect; however, it is also possible that the black square was simply distracting and it was this distraction (rather than the occlusion of the articulatory information) what caused the effect going away. I am not sure whether this can be fully addressed without an additional experiment, but the authors can at least acknowledge this possibility.

Thank you for raising this point. As laid out in the main part of the letter, the revised discussion of Experiment 2 now acknowledges and discusses this possibility.

Methods - > Exclusions:
-"We removed participants who […] (4) had unusually fast or slow reaction times" -> Please add more details as to what exact criteria where used here.
-"For this purpose, we considered participants with significant slopes in the opposite of the expected direction as likely having swapped the response keys" -> What about participants with flat slopes?

We have added the requested details (participants were excluded if their mean log reaction time was 3 standard deviations above or below the mean). Participants with flat slopes were included.

Related to this question, we have also added new plots to the SI that show by-participant and by-trial RTs, and how they relate to our exclusion criteria.

Figures 2 and 4: Please include in the legend information about all three lines.

We apologize for the confusion. The gray line actually showed the result of the audio-only (no pen) test continuum presented in Liu and Jaeger (2018, in their supplementary information). This gray line was only meant to show in our supplementary information, where we explain its relevance. This has now been fixed.

Overall, this is a very nice study that addresses an interesting question; however, I am afraid I cannot recommend it for publication given the issues identified above. I would be happy to look at a revision, if the authors are able to address the main two issues pointed out above.

Thank you again for your careful assessment of the work.

Reviewer 2

[summary omitted] I thought this was a clever study with exciting results (all the more reason I am sorry about my delayed review!) Compensatory processes are both broader than previously thought (in the sense that non-phonetic information leads to compensation, but also narrower than I might have assumed in that it has to be perceptually transparent information, and not inferred - which I initially found surprising, but makes sense since it is not a given that the presence of a pen will always lead to the same articulatory consequences). I also found the proposal at the end for why we might not see perceptual learning when a speaker has a pen in their mouth to be intriguing - if indeed this is due to compensation prior to any perceptual learning, this could indicate that higher level knowledge (i.e., I know this pen could be influencing the production, so I'll wait to see what happens without it) does not play the role we thought in perceptual learning (and would fit with other evidence that perceptual learning is heavily influenced by low-level factors like acoustic similarity).

Thank you for these kind words! And, yes, we were similarly intrigued by this possible explanation, which differs in some relevant ways from both of the explanations that have previously been advanced to explain to explain why we might not see perceptual learning when a speaker has a pen in the mouth (causal inference/'explaining way', Kraljic et al., 2008; Liu & Jaeger, 2018; vs. separate exemplar storage for audio-visual vs. audio-only exemplars, Kraljic & Samuel, 2011).

I don't have any concerns about the study itself or the conclusions. My main issue with the manuscript is that it is so tersely written that it omits important information and is sometimes confusing to follow. I really appreciate the extensive detail provided in the supplementary material (it's rare to get such a transparent walkthrough of the research process). However, I don't think a reader should have to refer so often to the supplemental materials to understand basic aspects of what is being presented in the main text. Some of this information belongs in the main text as well.

Agreed! R1 also requested more details on the methods to be included in the main part of the paper. And so we have moved many of the details from the SI into the paper. This includes clarifications for all of the questions raised below (thank you for listing them! That was very helpful in deciding what detail to include in the text).

Here are some of the questions/points of confusion I had:
-what does it mean to be a video in kind?

By "in kind", we meant a different video which fulfilled all the same criteria as the original (in this case, similar phonetic context and original fricative). We have clarified these criteria in the revised manuscript.

-participants were excluded if they 'had unusually fast or slow reaction times' - these criteria are not specified in the text.

The methods now clarify that participants were excluded if their mean log reaction time was 3 standard deviations above or below the mean.

-I'm not entirely clear on what is happening with the white dot catch trials - I had assumed that if a participant failed these trials (indicated there was a white dot when there wasn't) that they would be discarded. But instead it seems that only the trials themselves were discarded, not the participants?

None of the experiments actually had any trials with white dots. We only included the instructions about the white dots because we had planned to use such catch trials in subsequent perceptual recalibration experiments (which we have since done). This meant that the only way to make a mistake that related to catch trials was to press SPACE on trials that were actually *not* a catch trial.

As we now clarify, participants very *rarely* made that mistake (0.8% of responses in Exp 1a-c, and 0.3% of responses in Exp 2). However, since pressing the SPACE bar auto-advanced to the next trial, trials with SPACE bar responses are lacking categorization responses, and thus had to be excluded.

This comment also made us realize that while we had intended to additionally exclude individual trials whose reaction times were too fast or too slow, this exclusion was not run in our initial submission. The revised manuscript excludes some additional trials (<2% over all experiments) as these trials had within-participant scaled log-transformed RTs outside of 3 SD of mean of scaled log-transformed RTs at that trial position.

In- or excluding these trials does not change any of our results.

-it is impossible to interpret the figures without reading through the supplementary materials, because it is unclear what the extra lines on the figures refer to (results from a previous study).

We apologize for the confusion. Only our R markdown-based SI was meant to contain the comparison to Liu and Jaeger (2018, they gray line in the figure). This also confused another reviewer (understandably given that the legend didn't mention the gray line at all!).

We have now removed the gray line from the plot in the main text. In the SI, we include the comparison to Liu and Jaeger (2018), and clarify its relevance: comparison to the audio-only (and thus no pen) continuum used in Liu and Jaeger (and all previous work) helps to assess whether a audiovisual stimulus with the pen in the hand is actually identical to an audio-only stimulus (an *assumption* made in some previous work, e.g., Kraljic & Samuel, 2011, but not directly relevant to the present work).

-it is impossible to understand the difference between the a-c versions of Experiment 1 without reading the supplemental materials. All the main text says is that there were different test stimuli across their experiments, because they were chosen such that they elicited a certain percent of s and sh responses. But it's not clear why that would have changed so much across experiments until you see the issues that arose in the supplement (and it's so important to understand the differences because the predictions and a great deal of the discussion of Experiment 2 involve relating them to different conditions of Experiment 1 - for example, on p20 "Under this alternative hypothesis, both pen conditions (pen in mouth vs. hand) of Experiment 2 should yield rates of ashi-responses comparable to the pen in mouth condition in Experiment 1c (since Experiment 2 occludes most direct visual evidence of fricative articulation.)" All that needs to be said is that there were errors in labelling the continuum tokens and so the acoustic region covered by the continuum differed across these experiments, with Experiments 1a and 1b being shifted relative to Experiment 1c….

We have followed these suggestions (thank you for going the extra step of identifying the parts that are critical for understanding the experiments).

-"Here and for all other experiments, all 'main' effects were assessed in the center of the acoustic continuum for the first test block" - What counts as the center? Again, more specifics needed!

This was poorly worded: we meant the center/mean of the continuum steps (a direct consequence of centering and standardizing continuum as a predictor for our analysis). We have rephrased this.

Once again, I'm really enthusiastic about this work and believe it will have a big impact on our understanding of fundamental aspects of speech perception and learning.

Thank you!

Reviewer 3

[summary omitted] The paper describes a novel perceptual effect that will have wide interest. I would recommend publication after some revisions.

Thank you for these kind words!

I find the abstract to be short and missing some relevant information - namely, the overall finding. Please revise.

Thank you for catching this. We have expanded the abstract to describe the finding.

Pg. 15, Figure 2: this figure is challenging to parse. From the legend it looks like the gray line should be showing the effect of the pen in the hand across experiments, but that can't be true because the gray line is identical across panels. Please state what the gray line is indicating and also change the legend so the line is darker.

We apologize for the confusion. The gray line actually showed the result of the audio-only (no pen) test continuum presented in Liu and Jaeger (2018, in their supplementary information). This gray line was only meant to show in our supplementary information, where we explain its relevance. This has now been fixed.